

TEAM PROPOSAL

TEAM NAME	NEW MOON
UNIVERSITY	University of Kelaniya
PRODUCT NAME	Gesture Mouse AI
CATEGORY	AI / Computer Vision & Human-Computer Interaction

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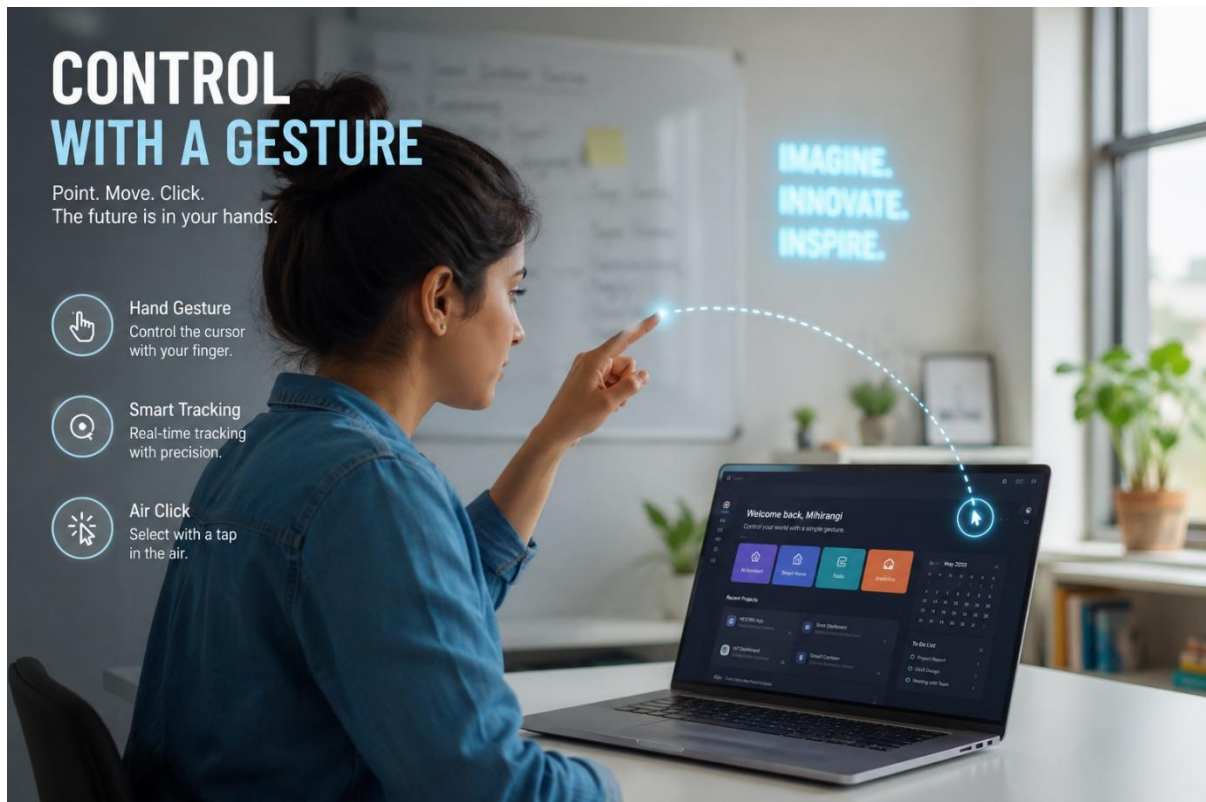
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1. Executive Summary

Gesture Mouse AI is a software-only, AI-powered touchless mouse that lets users control their computer using only a standard laptop webcam and natural hand gestures. Traditional input devices create accessibility barriers for users with limited hand mobility, hygiene concerns in shared environments such as hospitals and public kiosks, and friction in settings like presentations and smart classrooms. Existing touchless alternatives typically depend on costly external hardware such as depth cameras or infrared sensors, limiting their reach.

Our solution uses Google's MediaPipe hand-landmark model together with OpenCV to track hand movement in real time from an ordinary RGB webcam feed, and PyAutoGUI to translate recognized gestures into OS-level cursor and click actions. Moving the index finger moves the cursor, a pinch performs a left click, and additional gestures enable right-click, double-click, scroll, and drag-and-drop, all with adjustable sensitivity and jitter smoothing.

Our target users include people with limited fine-motor control, educators and presenters who need hands-free screen control, and institutions deploying public kiosks or smart classrooms where touch surfaces are undesirable. By requiring no additional hardware, Gesture Mouse AI is free to adopt, portable, and instantly deployable, offering a more accessible and cost-effective alternative to existing hardware-dependent gesture-control products. We plan to release it as a free downloadable application initially, with a premium tier for institutions later.



2. Problem Definition & Solution

2.1 Problem Definition

Traditional computer interaction relies heavily on physical input devices such as a mouse or trackpad. This creates real barriers for three overlapping groups: (1) users with limited hand mobility or physical disabilities who find fine motor control difficult; (2) people who must share devices in hygiene-sensitive environments such as hospitals, clinics, and public kiosks; and (3) presenters and educators in smart classrooms who need to control a screen from a distance without walking back to a keyboard or touchpad.

2.2 Problem Analysis

Growing demand for touchless and accessible interfaces has been accelerated by post-pandemic hygiene awareness and the broader rise of assistive technology. Existing gesture-control products, such as Leap Motion and other depth-sensor kits, demonstrate that there is real market interest in gesture-based interaction, but their reliance on specialized external hardware (depth cameras, infrared sensors) makes them expensive and inaccessible to the average student, educator, or small institution. This leaves a clear gap for a low-cost, software-only solution that runs on hardware people already own.

2.3 Proposed Solution

Gesture Mouse AI closes this gap using only the laptop's built-in camera. The system continuously captures the video feed, detects the user's hand using AI-based hand-landmark tracking, classifies the current gesture, and translates it into the corresponding mouse action in real time. Because no additional sensor or hardware purchase is required, the solution can be adopted instantly by anyone with a standard laptop, directly addressing the cost and accessibility barrier identified above.

3. Product Overview

3.1 Product Overview

Gesture Mouse AI is a lightweight desktop application built entirely on freely available computer-vision tools. It maps intuitive hand gestures to standard mouse operations:

- Moving the index finger — moves the mouse cursor
- Pinching thumb and index finger — performs a left click
- A distinct gesture (e.g. raising three fingers) — performs a right click
- Additional gestures — enable double-click, scroll, and drag-and-drop

The core vision is to make touchless computer interaction free, portable, and instantly deployable on hardware people already own, rather than something that requires a specialized purchase.

3.2 In-Scope & Out-of-Scope

In scope for this stage: cursor movement, left click, right click, double-click, scroll, drag-and-drop, jitter smoothing, and a sensitivity calibration step, all running on Windows/macOS laptops with a built-in or USB webcam.

Out of scope for this stage: multi-hand/multi-user gesture tracking, mobile platform support, voice-command integration, and enterprise device-management dashboards. These are noted as future roadmap items rather than current deliverables.

3.3 Uniqueness & Special Features

Unlike Leap Motion and other depth-sensor gesture kits, Gesture Mouse AI requires no external sensors or hardware only the built-in webcam already present on almost every laptop. This makes it free to adopt, portable, and instantly deployable. Combined with real-time performance, jitter-smoothing algorithms, and an adjustable-sensitivity calibration step, it offers a more accessible and cost-effective alternative to existing hardware-dependent gesture-control products.



4. Market Analysis & Business Strategy

4.1 Target Audience & Market Opportunity

Primary target users include.

- (1) individuals with limited hand mobility who find a physical mouse difficult to operate.
- (2) educators and presenters who need touch-free control of a screen during lectures or demonstrations.
- (3) public kiosk and smart-classroom operators who want to avoid shared touch surfaces.

Sri Lanka's universities, assistive-technology communities, and public-sector digitalization initiatives (e.g. smart classrooms funded by the Ministry of Education) represent an accessible early market, with clear expansion potential into hospitals, museums, and other public-facing kiosks.

4.2 Competitor Analysis

Leap Motion and similar depth-sensor gesture-control kits demonstrate that there is genuine demand for touchless interaction, but they require users to purchase dedicated external hardware, which raises cost and limits adoption outside specialized labs. Software-only alternatives on the market are typically limited to single gestures (e.g. only cursor movement) or lack robust click, scroll, and drag support. Gesture Mouse AI is positioned as a full-featured, hardware-free replacement for a physical mouse, differentiated by requiring zero extra hardware while still supporting the complete range of mouse actions.

4.3 Initial Go-to-Market Strategy

Leap Motion and similar depth-sensor gesture-control kits demonstrate that there is genuine demand for touchless interaction, but they require users to purchase dedicated external hardware, which raises cost and limits adoption outside specialized labs. Software-only alternatives on the market are typically limited to single gestures (e.g. only cursor movement) or lack robust click, scroll, and drag support. Gesture Mouse AI is positioned as a full-featured, hardware-free replacement for a physical mouse, differentiated by requiring zero extra hardware while still supporting the complete range of mouse actions.

5. Business Model & Growth Plan

5.1 Business Model Canvas

Key elements of the business model:

- **Customer Segments:** individuals with mobility challenges, educators/presenters, public kiosk and smart-classroom operators
- **Value Proposition:** free, hardware-free, instantly deployable touchless mouse control
- **Channels:** direct download, university/accessibility-community partnerships, social media demos
- **Customer Relationships:** community support, feedback-driven updates
- **Revenue Streams:** free core app; premium tier for advanced features and enterprise deployment
- **Key Activities:** gesture-recognition R&D, UX calibration, community engagement
- **Key Resources:** computer-vision pipeline (OpenCV, MediaPipe), development team, open-source tooling
- **Key Partners:** universities, accessibility organizations, smart-classroom/kiosk vendors
- **Cost Structure:** development time, hosting for downloads/updates, marketing and outreach

A visual Business Model Canvas diagram is recommended for the pitch deck to present these elements more clearly.

5.2 Partnerships & Funding Strategy

Potential strategic partners include university disability-support offices, accessibility-technology NGOs, and institutions running smart classrooms or public kiosks that could pilot the tool. In its early stages, the venture will be self-funded through the founding team and supported by hackathon/innovation grants (such as HackX and similar startup-challenge funding), with a premium enterprise tier planned as the primary future revenue source once core adoption is established.

6. Technical Overview & Implementation

6.1 Technical Overview

The system is implemented entirely in Python using the following components:

- OpenCV — captures and processes the live video feed from the built-in camera frame by frame
- MediaPipe — detects hand landmarks and tracks finger positions in real time
- Custom gesture-recognition logic — interprets landmark positions/distances to classify gestures (pointing, pinching, multi-finger poses)
- PyAutoGUI — translates recognized gestures into OS-level mouse actions (move, click, scroll, drag)

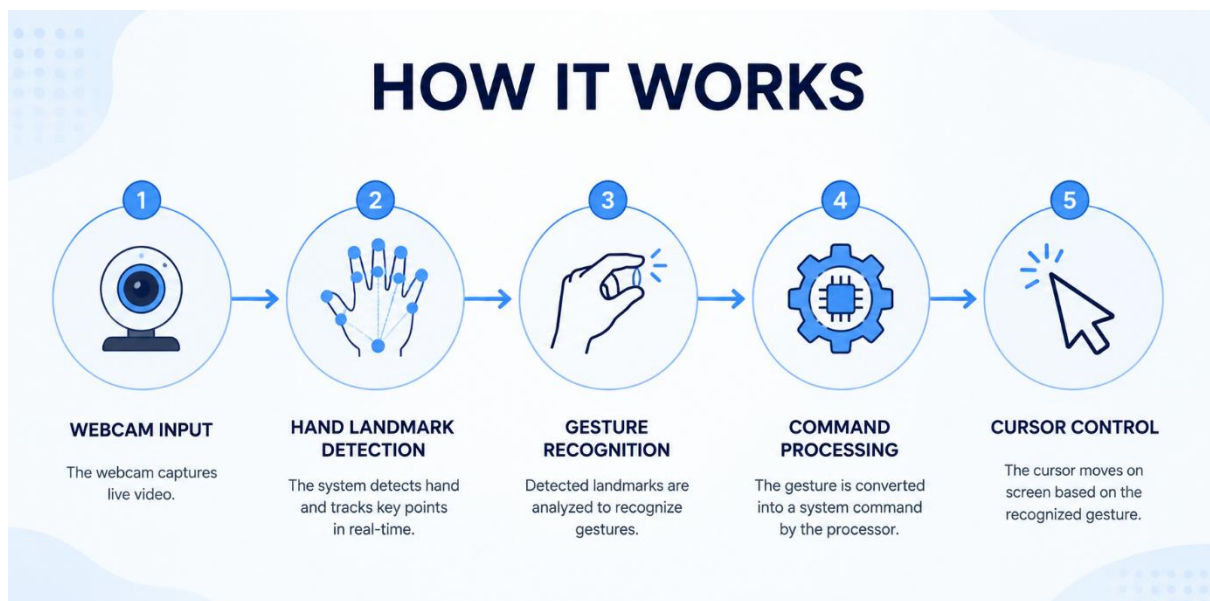
Pipeline: Camera Frame → Hand Detection (MediaPipe) → Landmark Extraction → Gesture Classification → Mouse Command Mapping (PyAutoGUI) → OS Cursor/Click Action.

6.2 Implementation Plan

Development will proceed in stages:

- (1) core hand-tracking and cursor-movement pipeline
- (2) click, right-click, and double-click gesture classification
- (3) scroll and drag-and-drop gestures
- (4) jitter-smoothing and sensitivity calibration
- (5) packaging as a lightweight installable desktop application
- (6) pilot testing with target users for feedback-driven refinement.

Major technical milestones include achieving stable low-latency tracking (near real time), a reliable gesture-classification accuracy target, and a smooth calibration flow that non-technical users can complete unassisted.



7. User Experience & Validation

7.1 User Scenario

Kasun, a university student with limited fine-motor control in his hand, struggles to operate a standard mouse for long periods. He opens Gesture Mouse AI on his laptop, and the built-in camera activates automatically. By moving his index finger, he moves the cursor to open his lecture slides; a simple pinch gesture clicks to advance the slide. During a group presentation, he uses the same gestures to control the screen from a distance, without touching the laptop, keeping the session smooth and hygienic in a shared classroom setting.



7.2 User Feedback (If Available)

Formal user testing has not yet been conducted. Our validation plan is to run a small pilot with classmates and lecturers (including at least one user with limited hand mobility) to gather structured feedback on gesture accuracy, comfort, and calibration ease before the next iteration, and to report results here once collected.

8. Roadmap

8.1 Product Roadmap

Phase 1 (MVP): core cursor movement and click gestures.

Phase 2: scroll, drag-and-drop, and sensitivity calibration.

Phase 3: multi-monitor support and custom gesture mapping.

Phase 4: enterprise features for kiosk/classroom deployment and centralized configuration.

8.2 Operational Roadmap

Phase 1 (MVP): core cursor movement and click gestures.

Phase 2: scroll, drag-and-drop, and sensitivity calibration.

Phase 3: multi-monitor support and custom gesture mapping.

Phase 4: enterprise features for kiosk/classroom deployment and centralized configuration.

8.3 Financial Roadmap

Early stages will run on minimal cost (development time, basic hosting), funded through the founding team and hackathon/innovation-grant funding. Revenue milestones will follow the introduction of a premium enterprise tier, with growth targets tied to the number of institutional pilots converted into paid deployments

9. Pitch Video

Link :- <https://youtu.be/F9Yy4gpLEBk?si=3HNRJborli19yi6s>

10. Team Vision & Information

Personal motivation from seeing accessibility challenges firsthand, the growing relevance of touchless interaction, and our team's shared interest in applied computer vision.

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